AI-Driven Sustainable Waste Analytics

Focus: SDG Goal 12 – Responsible Consumption & Production
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1. Problem Statement

Global waste generation is projected to reach **3.4 billion tonnes by 2050**, creating major environmental and infrastructure challenges.

This project addresses three key issues:

- **Infrastructure Gaps:** Identifying regions dependent on open dumping
- **Operational Inefficiency:** Replacing manual reporting with automated analytics
- **Predictive Deficiency:** Enabling proactive sustainability classification using AI

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2. Dataset Description

The dataset is derived from the **World Bank "What a Waste" database**.

Source: World Bank – What a Waste Global Database

- Coverage: Global data across countries
- Features include:
  - Waste generation (total & per capita)
  - Waste composition (plastic, food)
  - Treatment methods (recycling, dumping)

Final Dataset

After preprocessing:

```
216 records × 13 features
```

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3. Data Cleaning & Engineering

To ensure data quality and usability:

◆ Steps performed:

- **Column Standardization:** Cleaned and aligned inconsistent column names
- **Missing Value Handling:** Applied **median imputation** to preserve data
- **Type Conversion:** Converted all relevant columns to numeric format

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◆ Feature Engineering

A custom **Sustainability Index** was developed:

```
Sustainability Index =
(0.5 × Recycling Rate)
− (0.3 × Dump Rate)
− (0.2 × Plastic %)
```

- Recycling → positive impact
- Dumping → strong negative impact
- Plastic → moderate negative impact

◆ Normalization

The index was scaled between **0 and 1** for consistency.

◆ Target Variable Creation

To classify sustainability levels:

- Used **quantile-based binning (qcut)**
- Applied **ranking technique** to handle duplicate values

```
Classes:
Low | Medium | High
```

Final Distribution (Balanced)

```
Low       72
Medium    72
High      72
```

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4. Key Performance Indicators (KPIs)

The dashboard tracks:

1. **Total Waste Volume** – overall waste generated
2. **Average Recycling Rate** – efficiency of recycling systems
3. **Waste per Capita** – individual environmental impact
4. **Sustainability Index** – overall environmental score

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5. Exploratory Data Analysis (EDA)

🔍 Key Insights:

- **Plastic Waste Trend:** Higher-income regions generate more plastic waste
- **Urban Impact:** Strong correlation between urbanization and waste generation
- **Dumping Crisis:** Open dumping is dominant in low-income regions

☞ These insights support **data-driven environmental policies**

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6. Machine Learning Model

◆ Model Used:

Random Forest Classifier

◆ Why Random Forest?

- Handles non-linear relationships
- Reduces overfitting (ensemble method)
- Works well with structured datasets
- Provides feature importance insights

◆ Features Used

```
recycling_rate
plastic_pct
dump_rate
```

◆ Model

```
Accuracy: ~97% (due to target derived from input features)
```

Critical Insight:

The target variable (Sustainability Level) is derived from the same input features using a weighted formula.

Therefore:

- The high accuracy reflects the model's ability to **learn and replicate the defined relationship**
- It does **not represent independent real-world prediction performance**

Interpretation:

```
The model successfully captures sustainability patterns
based on environmental indicators.
```

◆ Feature Importance

1. **Recycling Rate** – most influential
2. **Dump Rate** – strong negative effect
3. **Plastic Percentage** – moderate impact

◆ Clustering (K-Means)

K-Means clustering was applied to group countries with similar waste profiles.

Purpose:

- Identify similar sustainability patterns
- Enable targeted regional strategies

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7. System Architecture & Deployment

◆ Flask Web Application

- Real-time sustainability prediction
- User inputs:
  - Recycling rate
  - Dump rate
  - Plastic %

👉 Output:

```
Low / Medium / High Sustainability
```

◆ Power BI Dashboard

An interactive dashboard was developed with:

- 🌍 **Global Map:** Waste distribution
- 📊 **Bar Charts:** Waste by region
- 📈 **Scatter Plot:** Recycling vs sustainability
- 💡 **KPIs:** Real-time performance metrics

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8. Business Insights & Recommendations

Key Recommendations:

- **Reduce Open Dumping:** Highest impact on improving sustainability
- **Increase Recycling:** Most influential positive factor
- **Target Plastic Waste:** Especially in high-income regions
- **Focus on Organic Waste:** Major component in developing regions

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9. Conclusion

This project demonstrates a complete **end-to-end AI solution**:

- Data Cleaning & Engineering
- Feature Design
- Machine Learning Model
- Dashboard Visualization
- Real-Time Deployment

It provides a **data-driven approach to sustainability decision-making**

This project demonstrates how data engineering, machine learning, and visualization can be combined to build a scalable, real-world sustainability analytics system.

Final Project Stack

- **Python:** Pandas, Scikit-learn, Matplotlib
- **Backend:** Flask
- **Frontend:** HTML/CSS
- **BI Tool:** Power BI
- **Model:** Random Forest + K-Means